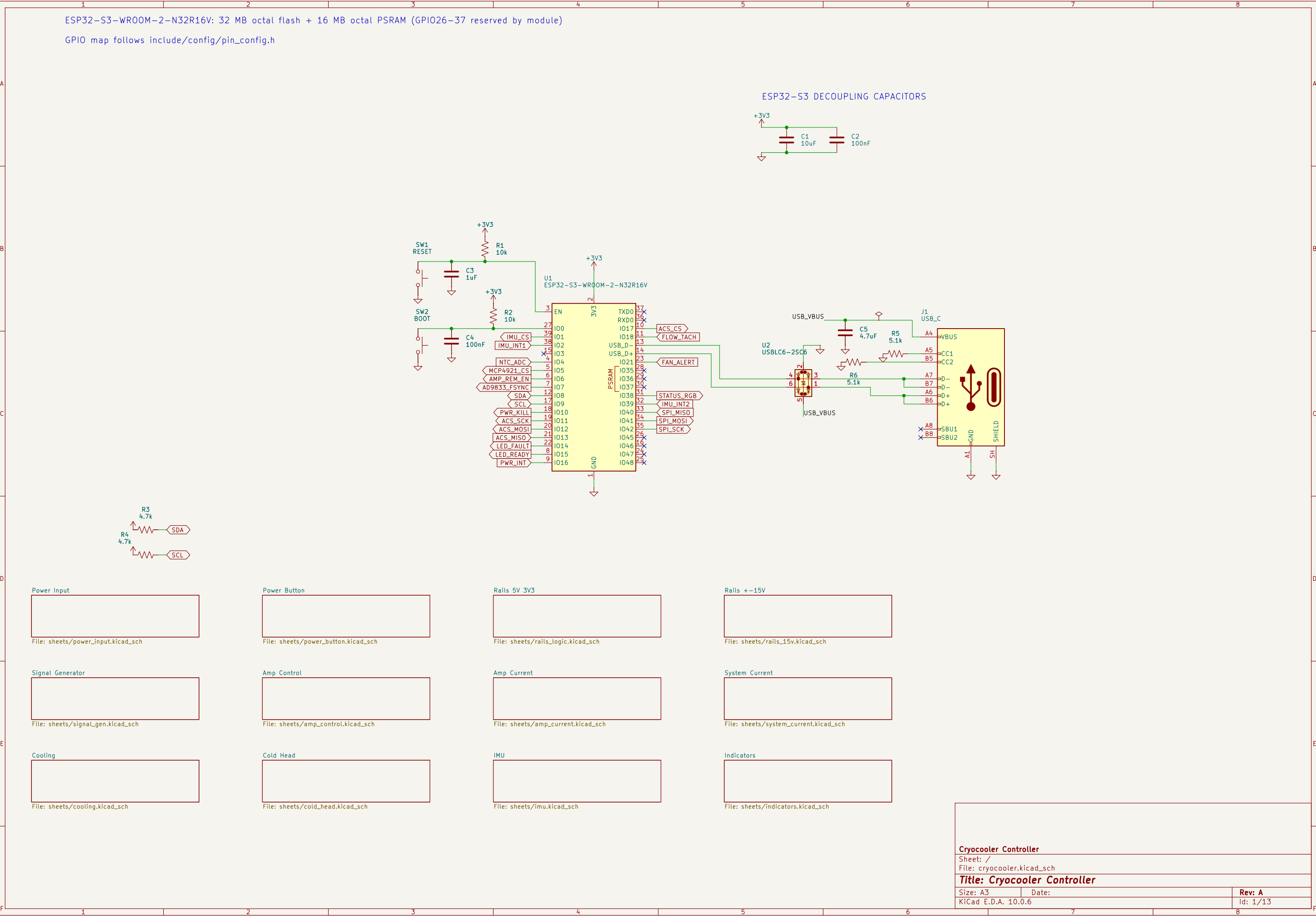
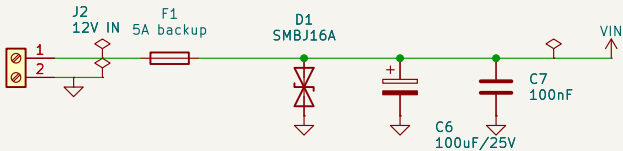




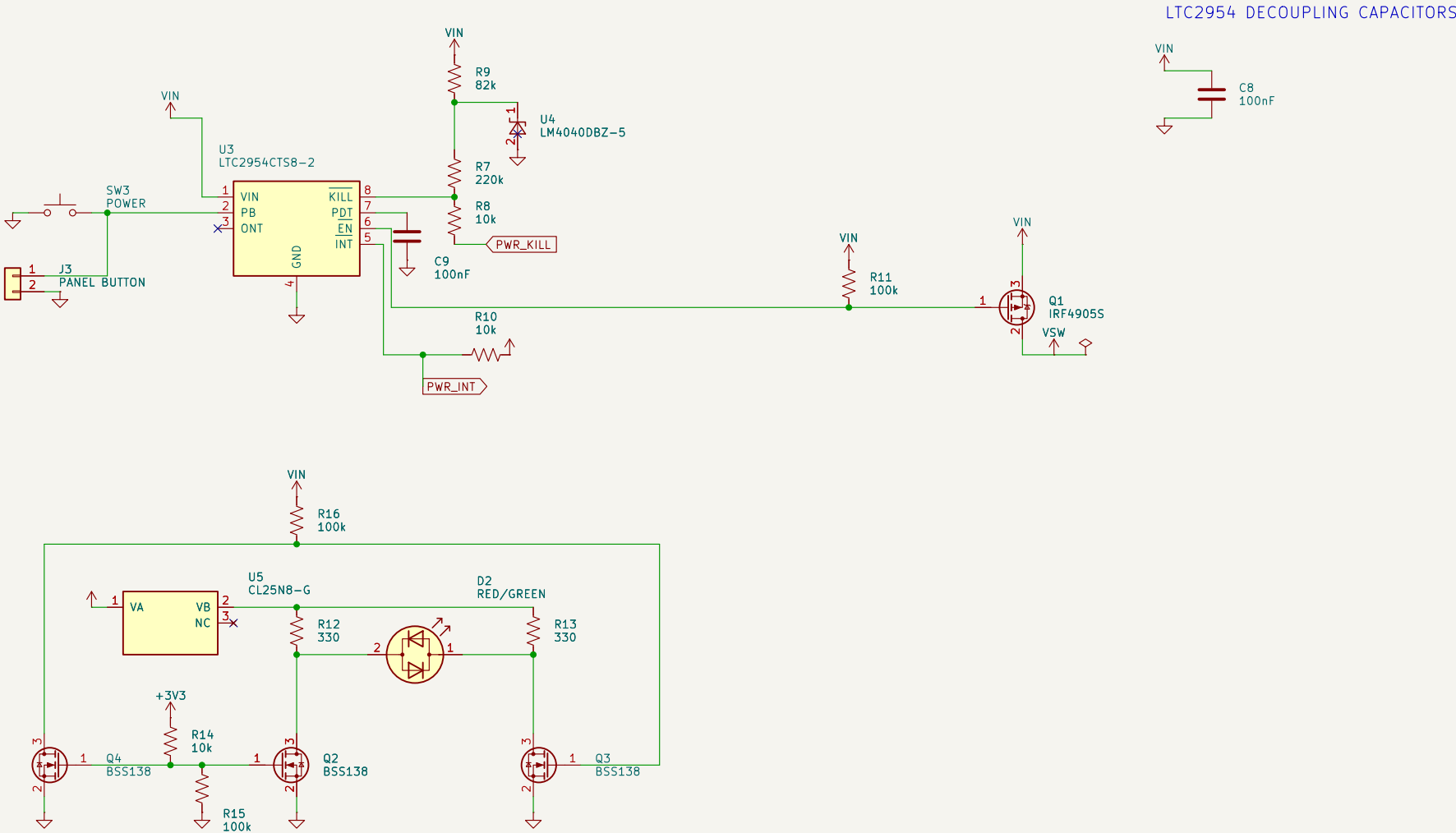
12 V input from a Dell XYK93 (DPS–2000EB class, ~12.2 V, ~160 A available): use the main 12 V rail with PS_ON held low.

F1 is a backup only: add an inline high–breaking–capacity fuse (7.5–10 A) at the PSU end. Reverse polarity: TVS forward–clamps and blows the fuse.



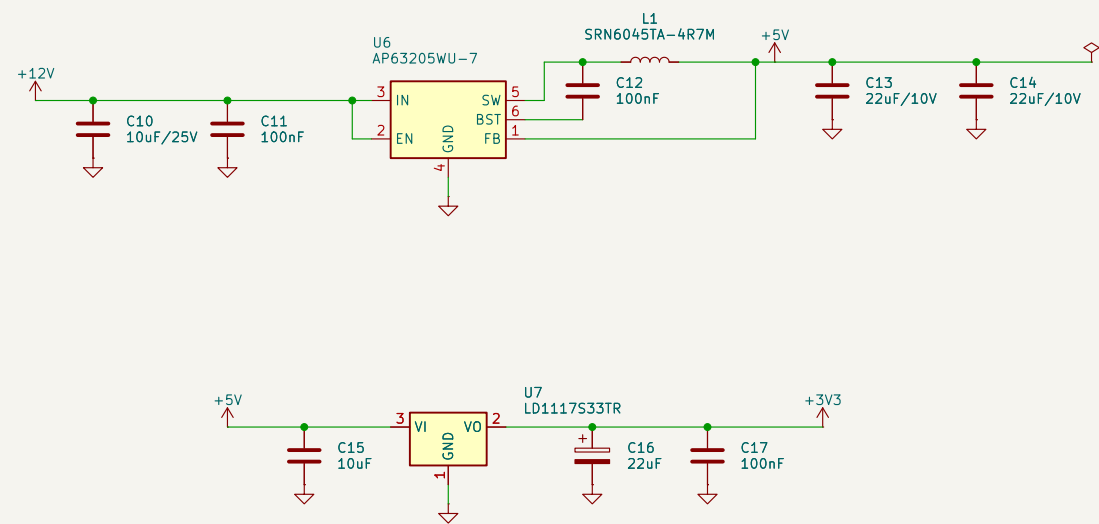
LTC2954-2: tap PB = on, hold = off. EN# drives the P-FET gate; KILL# from GPIO10 (active low, 100 ms pulse).

PWR_ON follows the +3V3 rail: on = green, standby = red. CL25 (25 mA) feeds both legs, ~9 mA reaches the LED, ~25 mA total from VIN in standby



12 V → 5 V (AP63205 buck, datasheet Fig. 1 / Table 3: 4.7 uH, 10 uF in, 2 x 22 uF out, 100 nF bootstrap) → 3.3 V (LD1117S33).

AP63205: VIN 3.8–32 V (40 V for 400 ms), EN tied to VIN, FB tied to the fixed 5 V output.



Cryocooler Controller

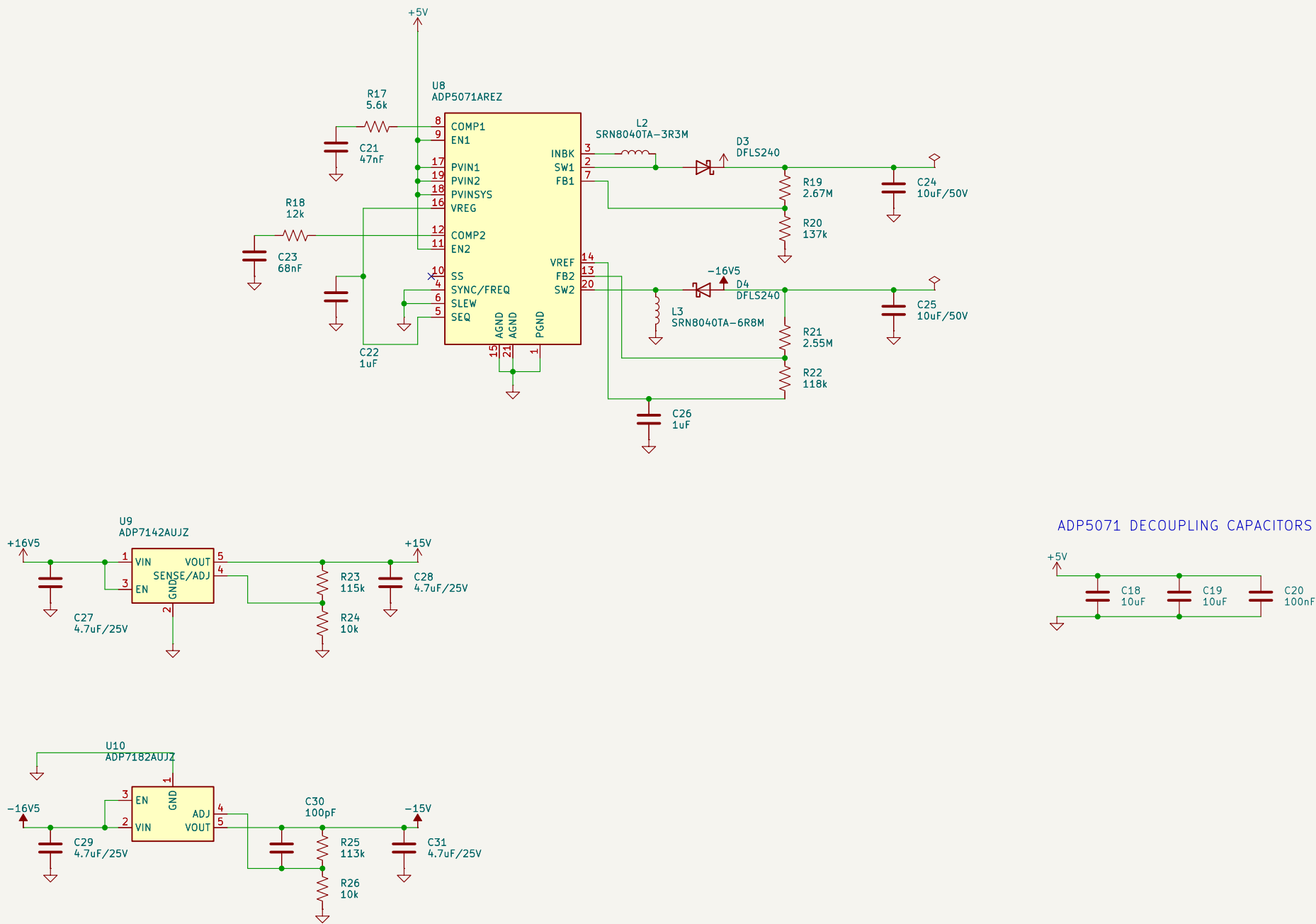
Sheet: /Rails 5V 3V3/
File: rails_logic.kicad_sch

Title: Rails 5V 3V3

Size: A3	Date:	Rev: A
KiCad E.D.A. 10.0.6		Id: 4/13

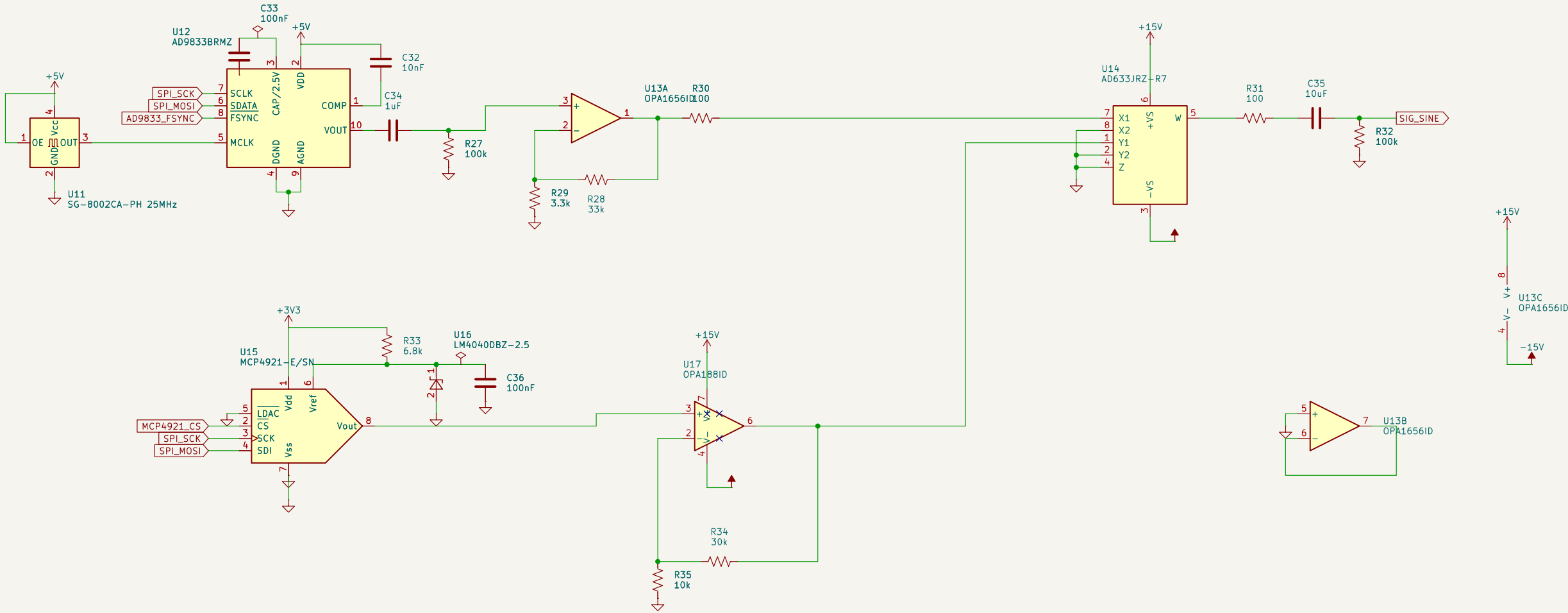
ADP5071AREZ (5 V in): +16.5 V boost and −16.5 V inverting rails; ADP7142 (+15 V) and ADP7182 (−15 V) post-regulate.

Values from ADP5071 datasheet Fig. 47 / Tables 9–11 (+/−16.5 V output rows rescaled via RFT). ADP7182 EN tied to VIN (auto-start); |VEN| >= 2 V enables.

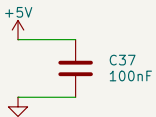


AD9833 60 Hz sine -> OPA1656 gain stage -> AD633 multiplier; MCP4921 + OPA188 give a 0-10 V amplitude control.

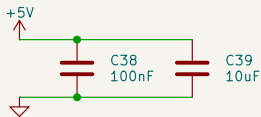
$V_{ctl} \text{ 0-10 V (2.5 V ref x4) } \times \text{ sine } \pm 3.3 \text{ V pk} / 10 = 0.33 \text{ V pk to amplifier input}$



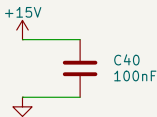
SG-8002CA DECOUPLING CAPACITORS



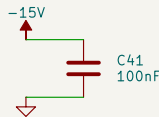
AD9833 DECOUPLING CAPACITORS



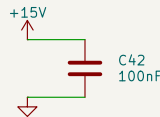
OPA1656 +15V DECOUPLING CAPACITORS



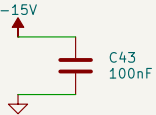
OPA1656 -15V DECOUPLING CAPACITORS



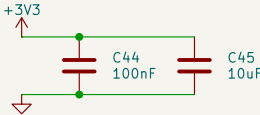
AD633 +15V DECOUPLING CAPACITORS



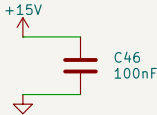
AD633 -15V DECOUPLING CAPACITORS



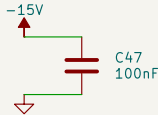
MCP4921 DECOUPLING CAPACITORS



OPA188 +15V DECOUPLING CAPACITORS



OPA188 -15V DECOUPLING CAPACITORS



Cryocooler Controller

Sheet: /Signal Generator/
File: signal_gen.kicad_sch

Title: Signal Generator

Size: A3

Date:

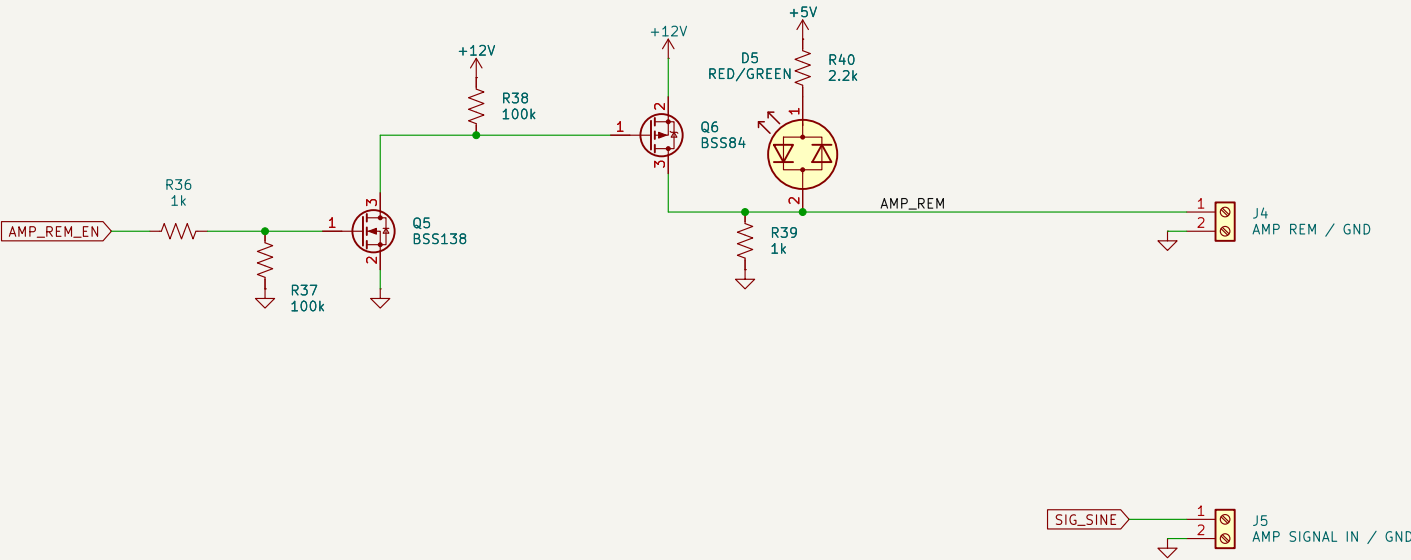
Rev: A

KiCad E.D.A. 10.0.6

Id: 6/13

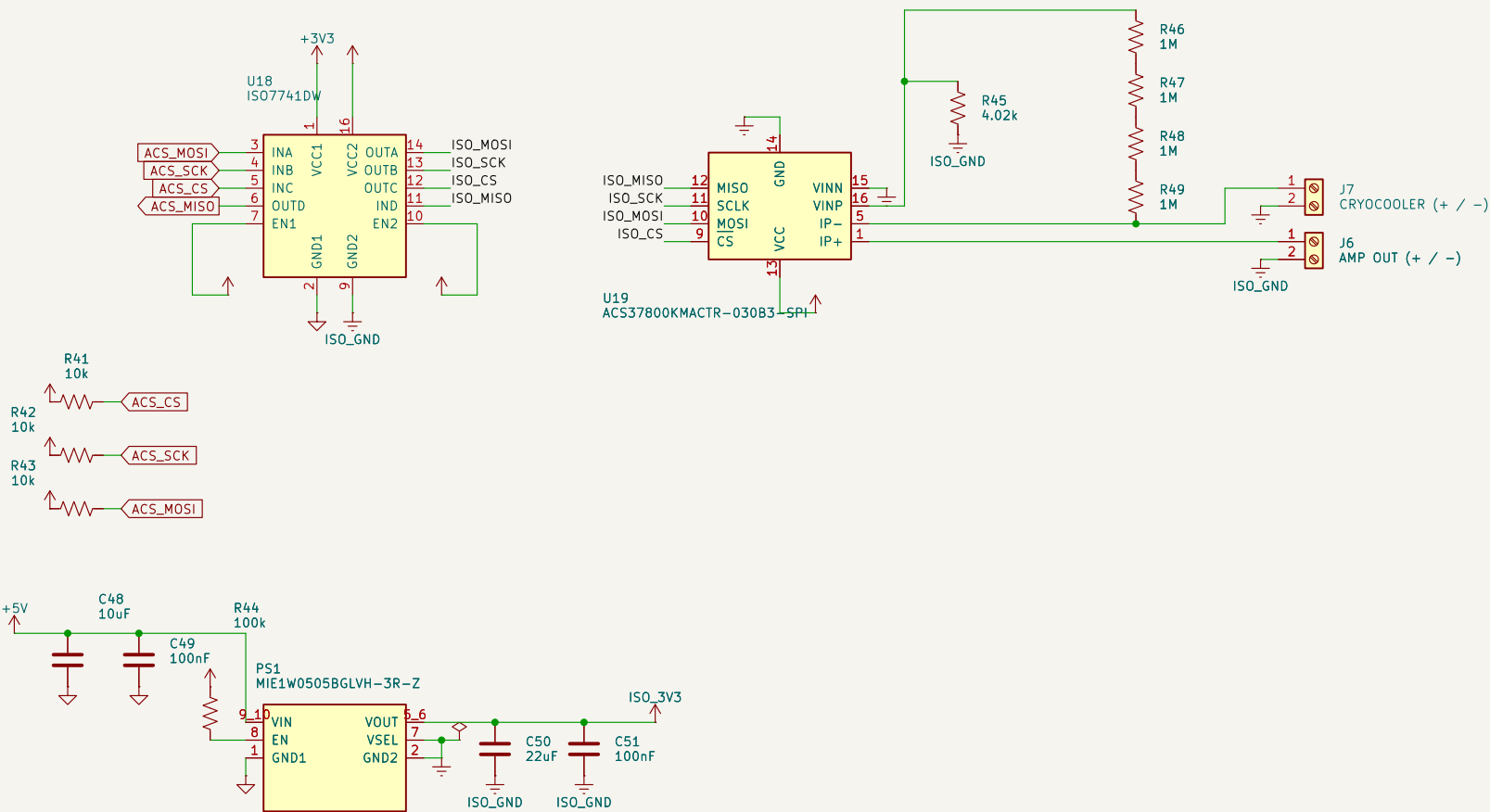
GPIO6 high -> +12 V on the amplifier REM terminal (P-FET high-side switch). LED: green = on, red = off.

Amplifier: AUDIOZERONE ZE500.1 (external): REM terminal + RCA/line input. Its power leads do not pass through this board.

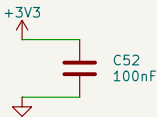


Amplifier output V/I monitor: ACS37800 (SPI) on an isolated 3.3 V domain (MIE1W0505, VSEL=GND2 -> 3.3 V); SPI bus GPIO11/12/13, CS GPIO17.

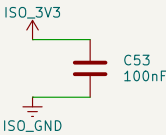
Mains-class creepage/clearance applies between ISO_GND domain and the logic side on the PCB.



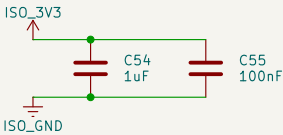
ISO7741 VCC1 DECOUPLING CAPACITORS



ISO7741 VCC2 DECOUPLING CAPACITORS



ACS37800 DECOUPLING CAPACITORS

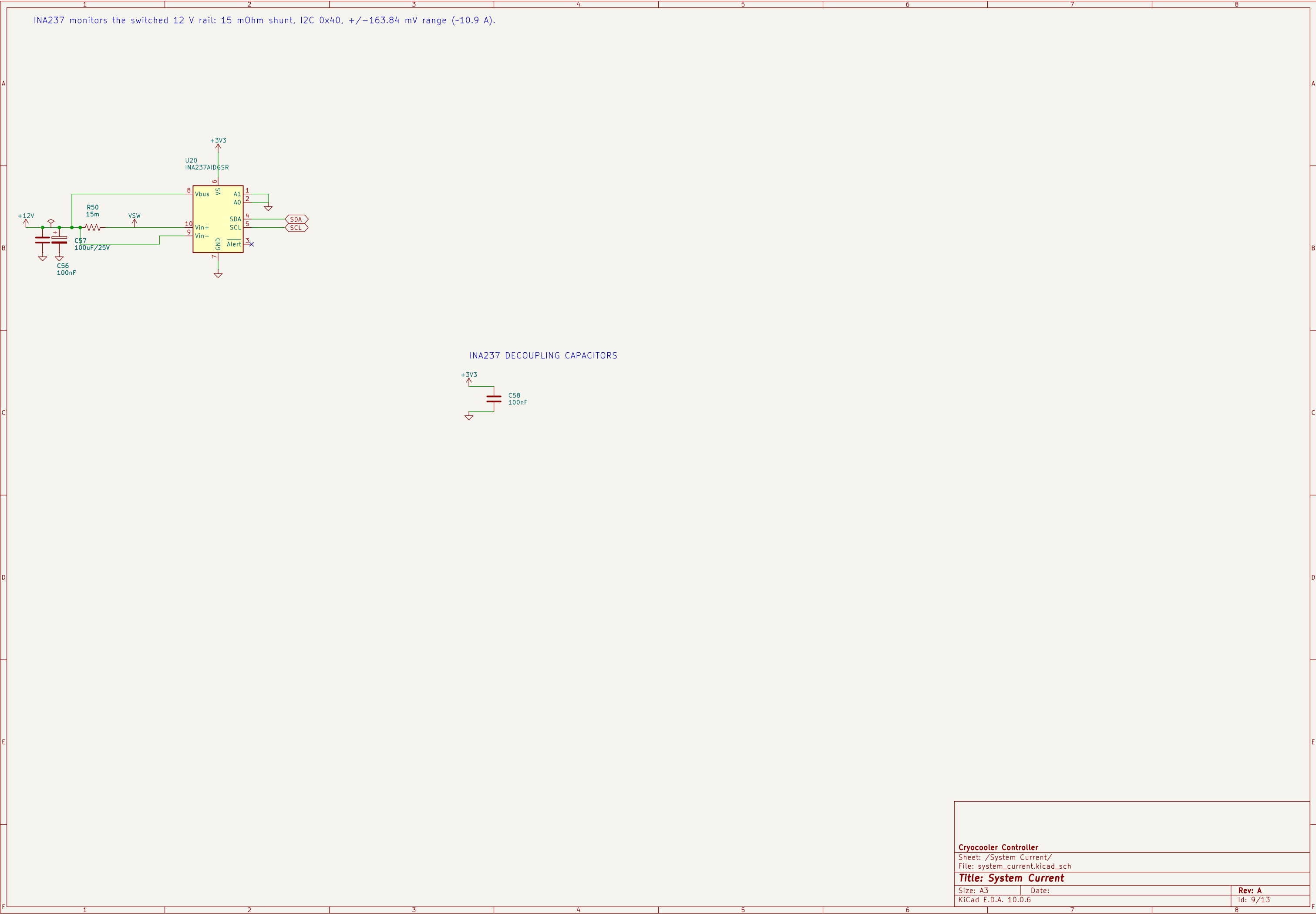


Cryocooler Controller

Sheet: /Amp Current/
File: amp_current.kicad_sch

Title: Amp Current

Size: A3	Date:	Rev: A
KiCad E.D.A. 10.0.6		Id: 8/13



EMC2303: ADDR_SEL 33k pull-up -> SMBus 0x4D (datasheet Table 5-1); CLK 4.7k pull-up -> default fan drive 0%. ch1 fans, ch2 pump. Alphacool ES: tach -> GPIO18, NTC -> GPIO4.

EMC2303 DECOUPLING CAPACITORS

EMC2303 MAIN CIRCUIT

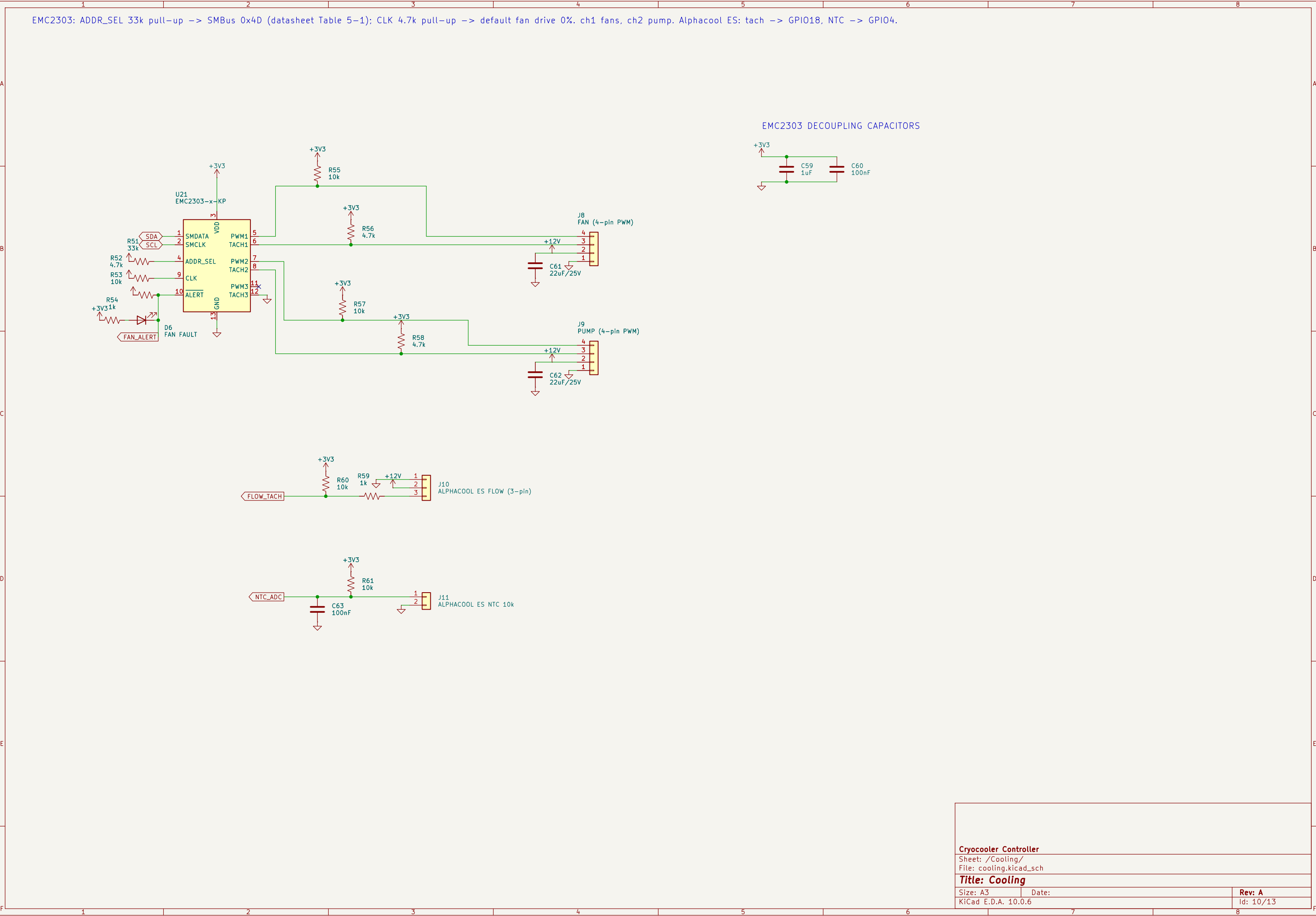
SENSORS

Cryocooler Controller
 Sheet: /Cooling/
 File: cooling.kicad_sch
Title: Cooling
 Size: A3 Date: Rev: A
 KiCad E.D.A. 10.0.6 Id: 10/13

EMC2303: ADDR_SEL 33k pull-up -> SMBus 0x4D (datasheet Table 5-1); CLK 4.7k pull-up -> default fan drive 0%. ch1 fans, ch2 pump. Alphacool ES: tach -> GPIO18, NTC -> GPIO4.

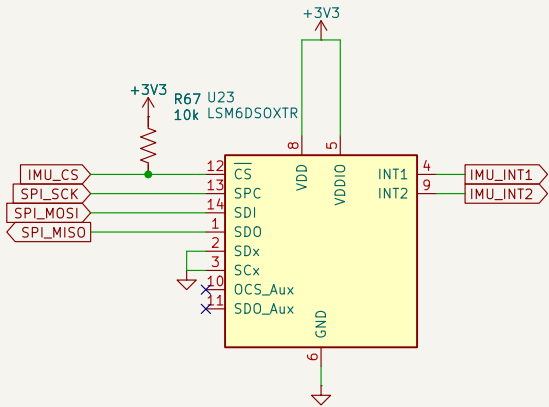
EMC2303 DECOUPLING CAPACITORS

Size: A3 Date: Rev: A
KiCad E.D.A. 10.0.6 Id: 10/13

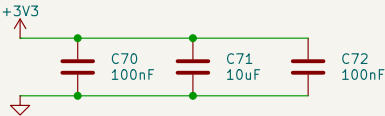


[illegible]

LSM6DSOX in 4-wire SPI mode on the main bus (CS GPIO1, INT1 GPIO2 single-tap, INT2 GPIO39 wake-up).



LSM6DSOX DECOUPLING CAPACITORS

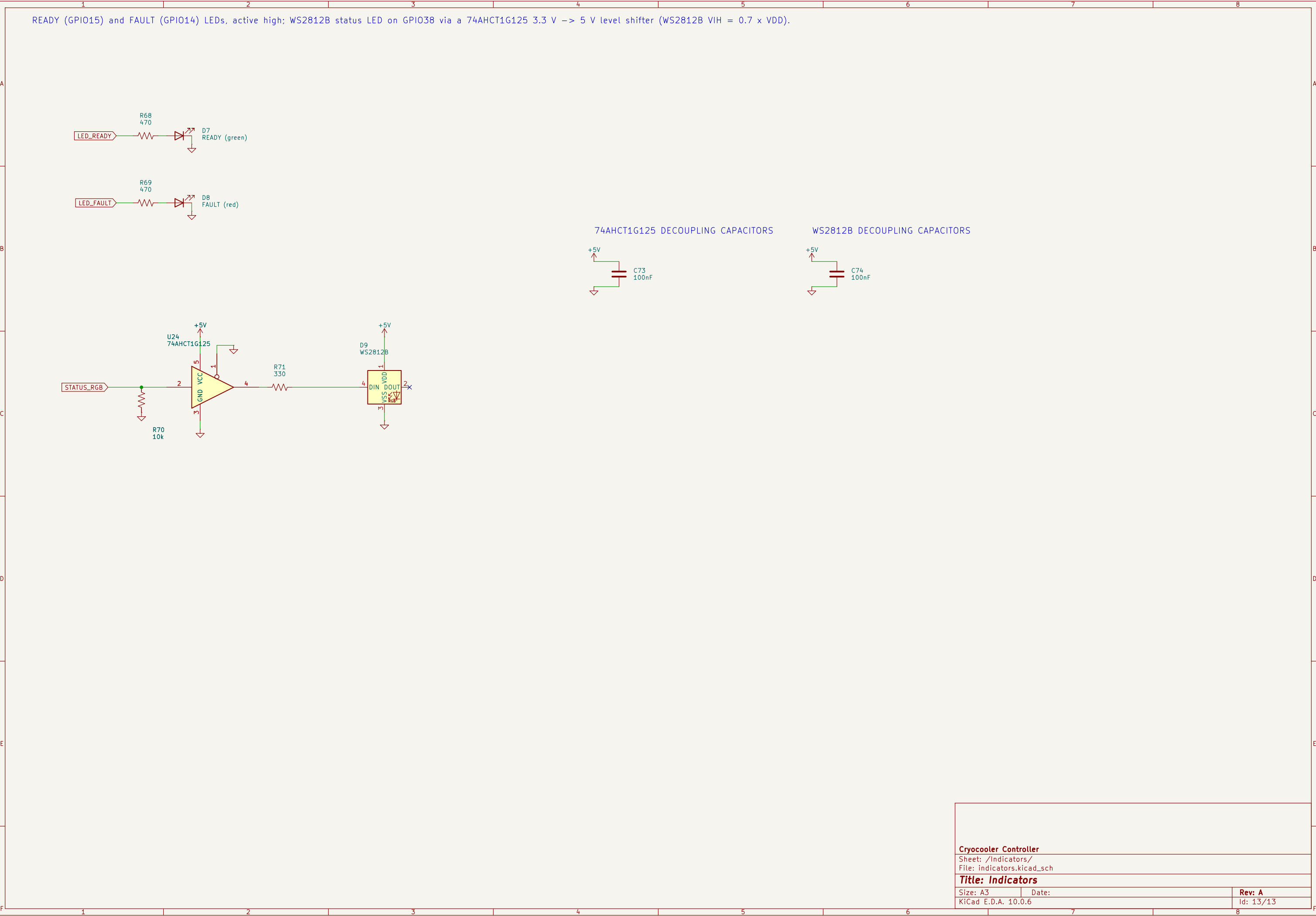


Cryocooler Controller

Sheet: /IMU/
File: imu.kicad_sch

Title: IMU

Size: A3	Date:	Rev: A
KiCad E.D.A. 10.0.6		Id: 12/13



Cryocooler Controller

Sheet: /Indicators/
File: indicators.kicad_sch

Title: Indicators

Size: A3	Date:	Rev: A
KiCad E.D.A. 10.0.6		Id: 13/13